

MXM Open House

Tuesday, December 9

Advertisement

- **Date:** December 9
- **Time:** 2-5 pm
- **Location:** Van Vleck 911
- Poster session to see results of undergraduate research this semester!
- Coffee & cookies

Math 421

Monday, December 8

Announcements

- This week's drop-in hours –
 - Wednesday 12:00-1:00 pm VV B205
 - Thursday 9:00 – 11:00 am VV B205
- Final Exam –
 - Date: Monday, December 15th
 - Time: 5:05-7:05 pm
 - Location: Bardeen 140
- Course Evaluations!
 - Open until 11:59pm on 12/10

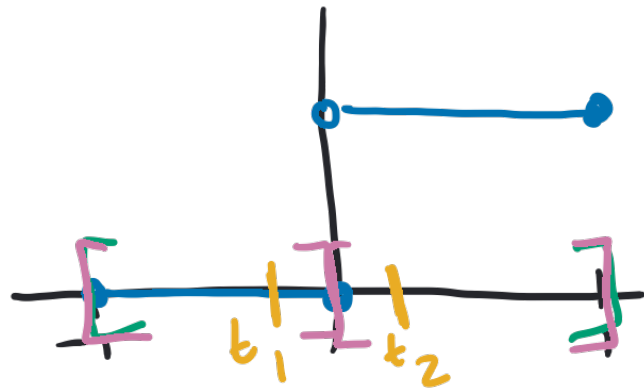
Equivalent Definition

if "inf of all
upper sums = sup
of all lower sums"

Theorem 32.5. ("Cauchy criterion for integrability") If f is bounded on $[a, b]$, then f is integrable on $[a, b]$ if and only if

for every $\epsilon > 0$, there exists a partition P of $[a, b]$
such that $U(f, P) - L(f, P) < \epsilon$

Example



- Show for every $\varepsilon > 0$ there exists $P = \{t_0 = a, t_1, t_2, \dots, t_n = b\}$ such that $U(f, P) - L(f, P) < \varepsilon$.

Let $f(x)$ be defined as follows. Prove f is integrable on $[-1, 1]$.

$$f(x) = \begin{cases} 0, & x \leq 0 \\ 1, & x > 0 \end{cases}$$

• $P_1 = \{-1, 1\}$

$$U(f, P_1) = 1(2) = 2$$

$$L(f, P_1) = 0(2) = 0$$

• $P_2 = \{-1, 0, 1\}$

$$U(f, P_2) = 0(1) + 1(1) = 1$$

$$L(f, P_2) = 0(1) + 0(1) = 0$$

$$P = \{-1, t_1, t_2, 1\}$$

$$0 \in [t_1, t_2]$$

$$U(f, P) = 0(t_1 - (-1)) + 1(t_2 - t_1) + 1(1 - t_2)$$

$$L(f, P) = 0(t_1 - (-1)) + 0(t_2 - t_1) + 1(1 - t_2)$$

$$U(f, P) - L(f, P) = t_2 - t_1 < \varepsilon$$

For example $t_2 = \frac{\varepsilon}{3}, t_1 = -\frac{\varepsilon}{3}$.

$$U(f, P) - L(f, P) = \sum_{k=1}^n (M_k - m_k) (t_k - t_{k-1})$$

Monotonicity $\Rightarrow$ Integrability

Theorem 33.1. Every monotonic function f on $[a, b]$ is integrable.

Proof: Assume f increasing.

$\rightarrow x_1 < x_2$ then $f(x_1) \leq f(x_2)$

If $f(a) = f(b)$, f is constant — proven integrable

1. Why is f bounded? $\forall x \in [a, b] \quad f(a) \leq f(x) \leq f(b)$
Assume $f(a) < f(b)$.

2. What does $U(f, P) - L(f, P)$ look like? $P = \{t_0, t_1, \dots, t_n\}$

$$U(f, P) - L(f, P) = \sum_{k=1}^n (f(t_k) - f(t_{k-1})) (t_k - t_{k-1}) < \epsilon$$

$$< \sum_{k=1}^n (f(t_k) - f(t_{k-1})) \cdot C = C \sum_{k=1}^n (f(t_k) - f(t_{k-1})) = C \cdot [f(t_1) - f(a) + f(t_2) - f(t_1) + \dots + f(b) - f(t_{n-1})]$$

3. What restriction should you impose on the length of each subinterval so that $U(f, P) - L(f, P) < \epsilon$?

$$U(f, P) - L(f, P) < C \cdot (f(b) - f(a))$$

$$\text{Let } C = \frac{\epsilon}{2(f(b) - f(a))}$$

$$P = \{t_0, t_1, \dots, t_n\} \text{ such that } t_{k-1} - t_k < \frac{\epsilon}{2(f(b) - f(a))}.$$

Continuity $\Rightarrow$ Integrability

Theorem 33.2. Every continuous function f on $[a, b]$ is integrable.

Proof Ingredients:

Bounded? $\checkmark$ Thm proven in class.

EVT - on each $[t_{k-1}, t_k]$ there exist x_k, y_k s.t. $f(x_k) = M_k$ &
 $f(y_k) = m_k$

Uniform continuity: $|t_k - t_{k-1}| < \delta$ then $|f(x_k) - f(y_k)| < \square$

$$U(f, P) - L(f, P) = \sum_{k=1}^n (M_k - m_k) (t_k - t_{k-1})$$

$$= \sum_{k=1}^n (f(x_k) - f(y_k)) (t_k - t_{k-1})$$

$$< \sum_{k=1}^n \square (t_k - t_{k-1})$$